

CRS-R Change Interpretation: Quick Reference

Using serial Coma Recovery Scale-Revised (CRS-R) assessments to interpret change over time

1 Start with the right scale for quantifying change

Raw CRS-R total (0-23)	Transformed measure (0-100)
Ordinal: useful for describing observed performance, but equal raw-score differences do not necessarily represent equal amounts of change.	Equal-interval: better suited for quantifying differences over time.

Example: a 1-point raw-score difference can represent very different amounts of change depending on where the scores fall on the CRS-R.

2 Ask whether the difference exceeds measurement error

Approach	What it does	Key consideration
MDC90 9 units	Applies one global threshold across the transformed measure.	90% confidence; less conservative than MDC95.
MDC95 11 units	Applies one global threshold across the transformed measure.	95% confidence; more conservative than MDC90.
cMDC95 pair-specific	Uses a threshold based on the specific scores being compared.	Accounts for differences in measurement precision across the measure.

Why conditional? Measurement error is not necessarily constant across a measure. Like many assessments, the CRS-R provides different levels of measurement precision at different points along the scale, with greater measurement error at the low and high extremes.

3 Interpret the result

Finding	Clinical interpretation
Improvement beyond error	The increase exceeds the selected measurement-error threshold, supporting interpretation of true improvement in neurobehavioral performance.
Within measurement error	The observed difference does not exceed the selected threshold. Current performance may not be measurably different from the prior assessment.
Decline beyond error	The decrease exceeds the selected measurement-error threshold, supporting interpretation of true decline. Consider assessment conditions and relevant medical or neurologic factors.

Use both lenses. State-of-consciousness transitions (Coma -> UWS/VS -> MCS- -> MCS+ -> eMCS) remain clinically essential. A patient may cross a diagnostic boundary with little movement in the total score or demonstrate measurable change within a diagnostic category.

Detectable != meaningful. Change beyond measurement error provides confidence that an observed difference reflects **true change rather than expected measurement variability**. It does not, by itself, tell us whether that change is meaningful to the patient or care partner.

Suggested interpretation language

"This patient demonstrated a 15-unit increase on the transformed 0-100 CRS-R measure, exceeding the selected measurement-error threshold. This indicates **improvement beyond measurement error** at the selected confidence level."

Remember: Interpret total-score change alongside CRS-R subscale findings, diagnostic state, assessment conditions, serial performance, and the broader clinical picture.

Sources: Weaver JA, et al. *J Neurotrauma*. 2024;41:e1996-e2008. Grady-Dominguez P, et al. *Arch Phys Med Rehabil*. 2026. Educational prototype for team review; not a substitute for clinical judgment.